

CT-Solve

1. Difference between Computer Architecture and Computer Organization.

Computer Architecture	Computer Organization
• Architecture describes what the computer does	• organization describes how it is done
• Architecture deals with the functional behavior of computer system	• Computer organization deals with the structural relationship.
• As a Programmer, you can view Architecture as a series of instruction Addressing modes, registers	• The implementation of the Architecture called organization
• Architecture focuses on its hardware	• organization focuses on its Performance

* which type of instruction set architecture is used in modern processors such as Intel core i7 - RISC, CISC, or a combination of both?

Ans: Modern core i7 processors are based on CISC Architecture - specially x86/x86-64 family. This Architecture provides a large and

complex set of instructions that allow a single instruction to perform multiple operations, making it

However, internally, they use RISC like micro-operations for execution. Inside

The processor, complex CISC instructions are decoded into smaller and smaller (uops). These micro-operations are then executed by the CPU's internal execution unit more efficiently.

This hybrid approach allows modern processors to maintain compatibility with the x86 architecture.

How it works:

- The processor receives complex CISC instruction (move data, multiply, etc)

- Inside the CPU, these are decoded into simple style micro operations

- These micro operations are then executed by a RISC-like core for speed and efficiency.

Core is -> Entry level
Core is -> Balanced performance
Core is -> Gaming and creative
Core is -> Enthusiast & high end performance

3.

a. RISC Processor has easily decodable Instruction format, so decoding is hard in CISC.

b. NO, Intel Processor will not fit into an AMD motherboard;

c. The four basic step of CPU operation

- Fetch

- Decode

- execute

- store

d.

Core i3 — Entry level

Core i5 — Balanced performance

Core i7, → Gaming and creative

Core i9 → Enthusiast & high end performance.

CT-2

MB = mega Byte
mb = mega bit

1. Let's say you're uploading a 2 gb video file with Transfer rate (R) = 50 mbps. Average access latency (TA) = 120 ms for time for handshake, connection setup., the calculate average time to read and write.

$$\therefore T_n = T_A + \frac{n}{R}$$

$$= 0.12 + \frac{2 \times 10^9 \times 8}{50 \times 10^6}$$

$$= 0.12 + 320$$

$$\approx 320.12$$

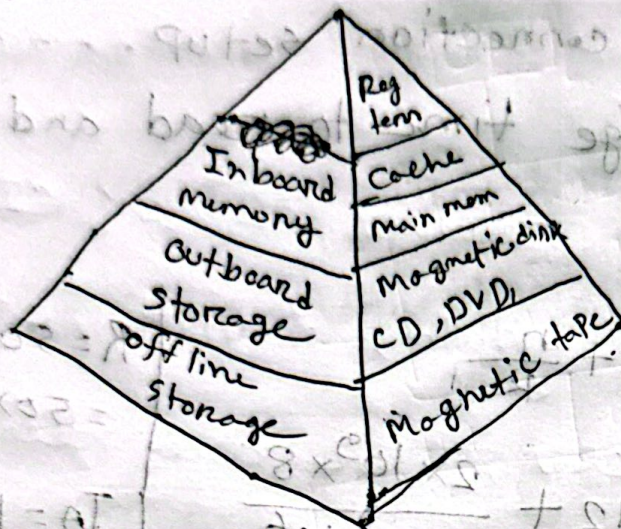
$$R = 50 \text{ Mbps} \\ = 50 \times 10^6 \text{ bit/s}$$

$$T_A = 120 \text{ ms} \\ = 0.12 \text{ s}$$

$$n = 2 \text{ GB} \\ = 2 \times 10^9 \text{ bytes} \\ = 2 \times 10^9 \times 8 \text{ bit}$$

2. Show the memory hierarchy and draw the single and three-level cache organization.

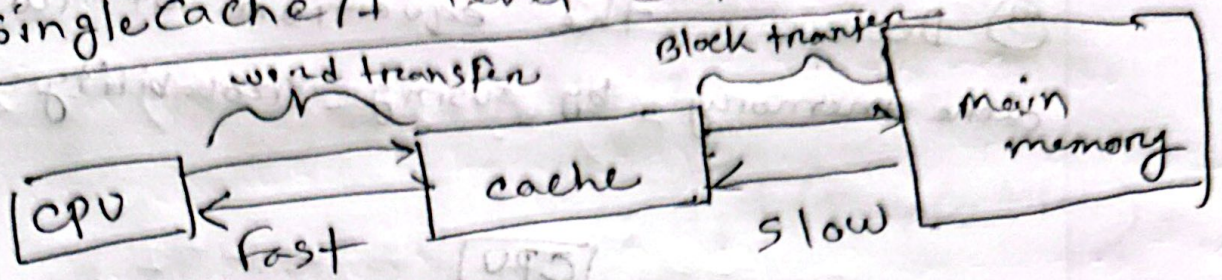
memory hierarchy:



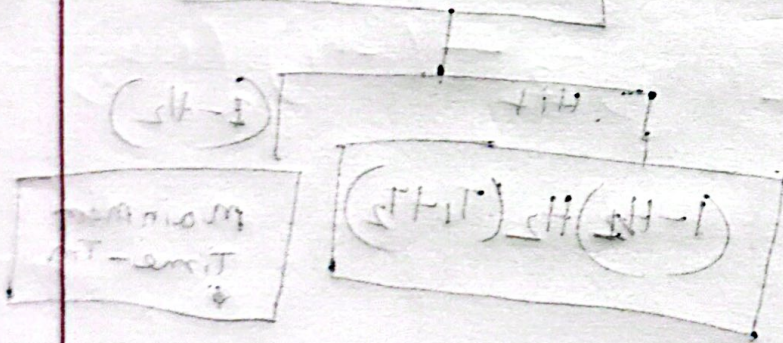
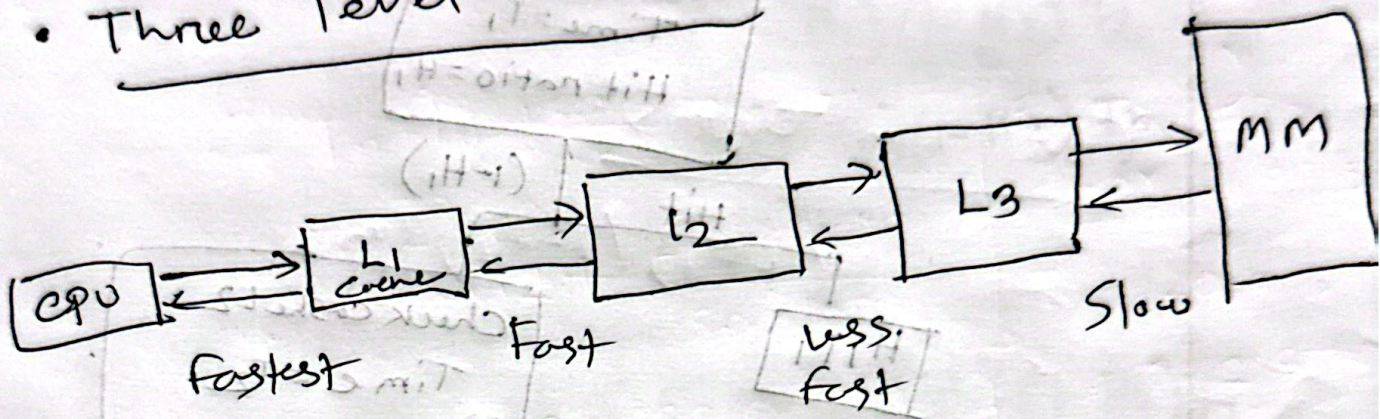
As one goes down, the following occur

- Decreasing cost per bit
- Increasing capacity
- Decreasing Access time
- Decreasing frequency of the memory by the processor

• Single cache / 1 level cache:

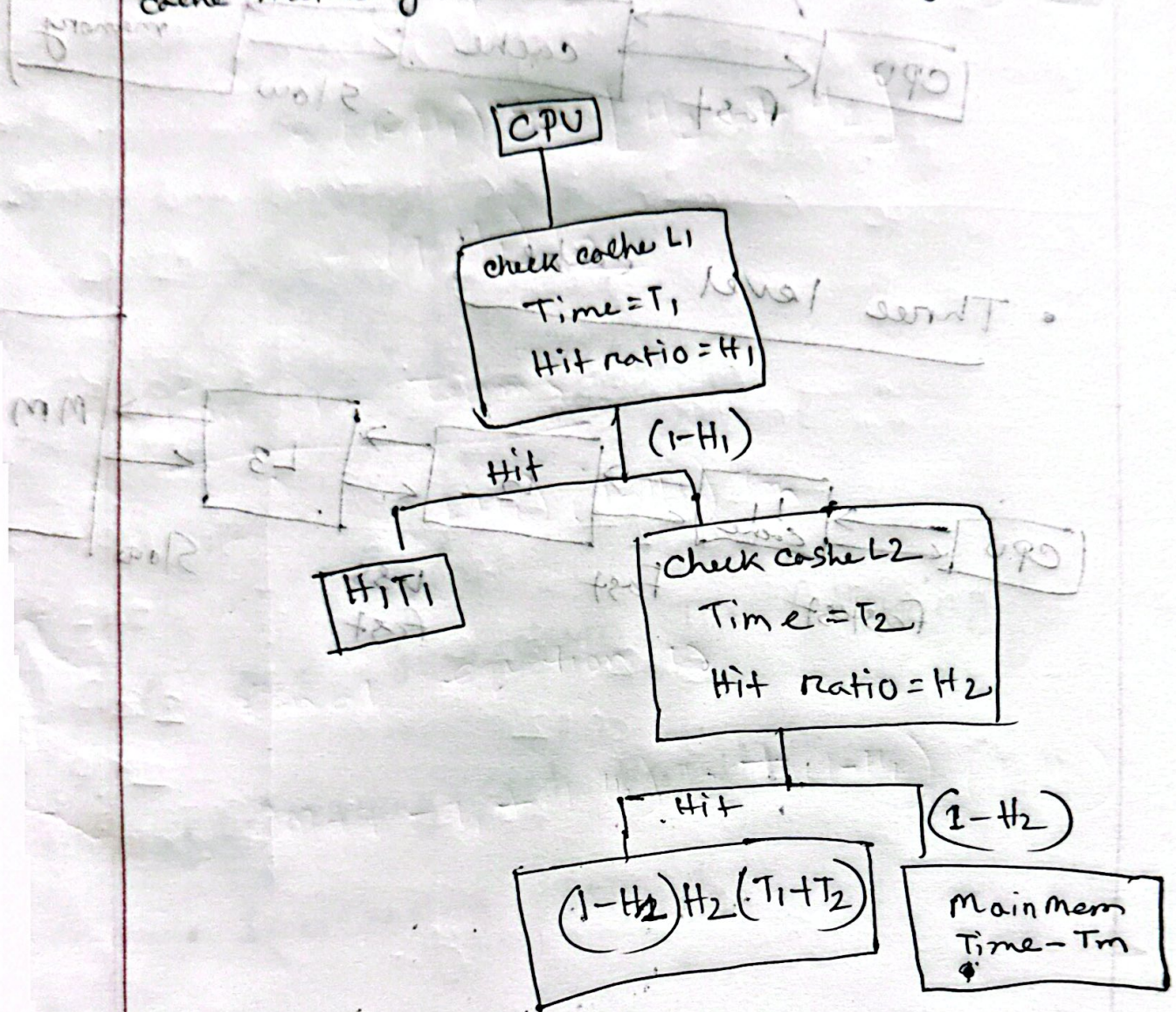


• Three level cache:



$$(L_1 - 1) \cdot (L_2 - 1) \cdot (L_3 - 1)$$

③ Derive the equation of two level cache memory by using Probability flowchart



$$(1-H_1) (1-H_2) (T_1+T_2+T_m)$$

2 level equation:

$$\begin{aligned} \frac{\sigma T_{AMAT}^4}{\sigma T_{eff}^4} &= H_1 T_1^4 + (1-H_1) H_2 (T_1 + T_2)^4 + \\ &\quad (1-H_1)(1-H_2) (T_1 + T_2 + T_m)^4 \\ &= H_1 T_1^4 + (1-H_1) H_2 T_1^4 + (1-H_1) H_2 T_2^4 + \\ &\quad (1-H_1)(1-H_2) T_1^4 + (1-H_1)(1-H_2) T_2^4 \\ &\quad + (1-H_1)(1-H_2) T_m^4 \end{aligned}$$

Now for T_1 terms:

$$\begin{aligned} &H_1 T_1^4 + (1-H_1) H_2 T_1^4 + (1-H_1)(1-H_2) T_1^4 \\ &= T_1^4 [H_1 + (1-H_1) H_2 + (1-H_1)(1-H_2)] \\ &= T_1^4 [H_1 + (1-H_1) (H_2 + 1 - H_2)] \\ &= T_1^4 [H_1 + 1 - H_1] \\ &= T_1^4 \end{aligned}$$

For T_2 terms:

no other level 2

$$+ (1-H_1)(H_2 T_2) + (1-H_1)(1-H_2) T_2$$

$$= (1-H_1) T_2 [H_2 + 1 - H_2]$$

$$= (1-H_1) T_2$$

For T_m terms:

$$(1-H_1)(1-H_2) T_m$$

IT not work

$$(1-H_1)(1-H_2) T_m$$

Final equation is

$$AMAT = T_1 + (1-H_1) T_2 + (1-H_1)(1-H_2) T_m$$

$$[(1-H_1) + (1-H_1)] T_2 =$$

$$[1-H_1 + 1-H_1] T_2 =$$

$$1-H_1$$